

Advanced Enterprise Network

Mixed Routing Design

Comprehensive network architecture featuring 5 managed switches, 8 workstations, 3 clients, and 6 routers with dual routing protocols, 3 north routers using static routing and 3 right routers implementing OSPF for optimal performance and reliability

Digital Egypt Pioneer Initiative (DEPI) Graduation Project

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Project Overview

This project details the design and configuration of a multi-department enterprise network featuring a mixed routing architecture, comprehensive VLAN segmentation, and essential network services. The implementation uses Huawei routers and switches within the eNSP simulation environment.

Devices: 6 routers, 5 switches, 8 PCs, 3 clients, 2 APs and 2 servers.

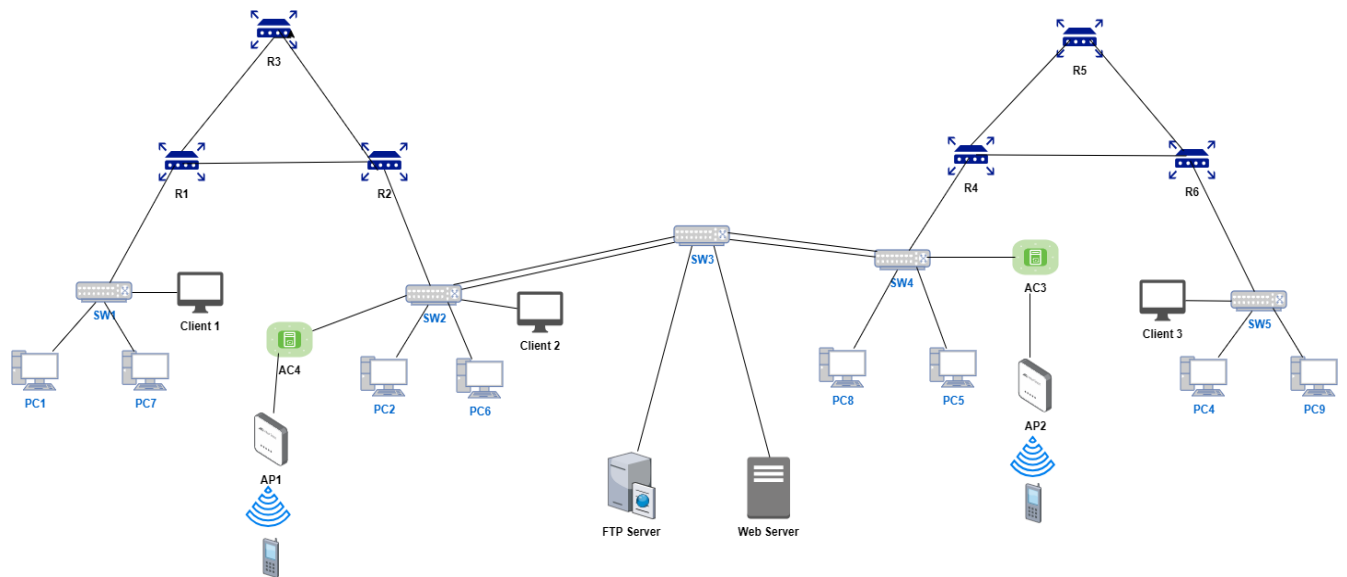
Features implemented: VLANs, static routing, OSPF routing, ACLs, Access Points, DHCP & FTP servers

Key Features Implemented

Feature	Description
Network Architecture	Hierarchical design with 6 routers, 5 switches, 8 PCs, 3 clients, 2 Access Points (APs), and 2 servers.
VLAN Segmentation	Five distinct VLANs (10, 20, 30, 40, 50) for traffic isolation and security.
Mixed Routing	Static Routing for the North Routers (AR1, AR2, AR3) and OSPF for the South Routers (AR4, AR5, AR6) to provide both predictable paths and dynamic adaptability.
First Hop Redundancy	VRRP (Virtual Router Redundancy Protocol) implemented on the Distribution Layer switches (LSW1, LSW2, LSW4, LSW5) to ensure high availability for VLAN gateways.
Network Services	FTP Server, Web Server, and Wireless Access Control (AC/AP).

Network Topology Overview

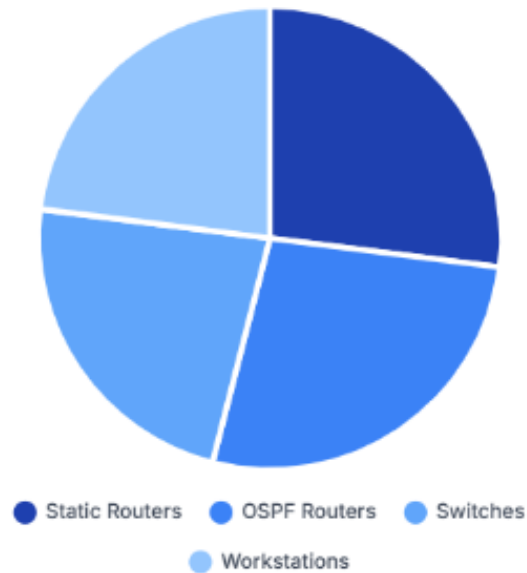
The network is structured into three main layers: Core (Routers), Distribution (Switches), and Access (End Devices)



- Six routers configured with dual routing protocols for optimal performance
- Three north routers implementing static routing for predictable paths
- Three right routers using OSPF for dynamic adaptive routing
- Five managed switches providing segmented connectivity for eight workstations

This hybrid network design combines the reliability of static routing with the adaptability of OSPF protocols, providing both predictable traffic flows and dynamic network resilience for enterprise operations.

The mixed routing approach ensures optimal performance while maintaining network stability and providing automatic failover capabilities for critical business operations.



Device Distribution

Network infrastructure allocation across routing domains and access layers

Network Topology Description

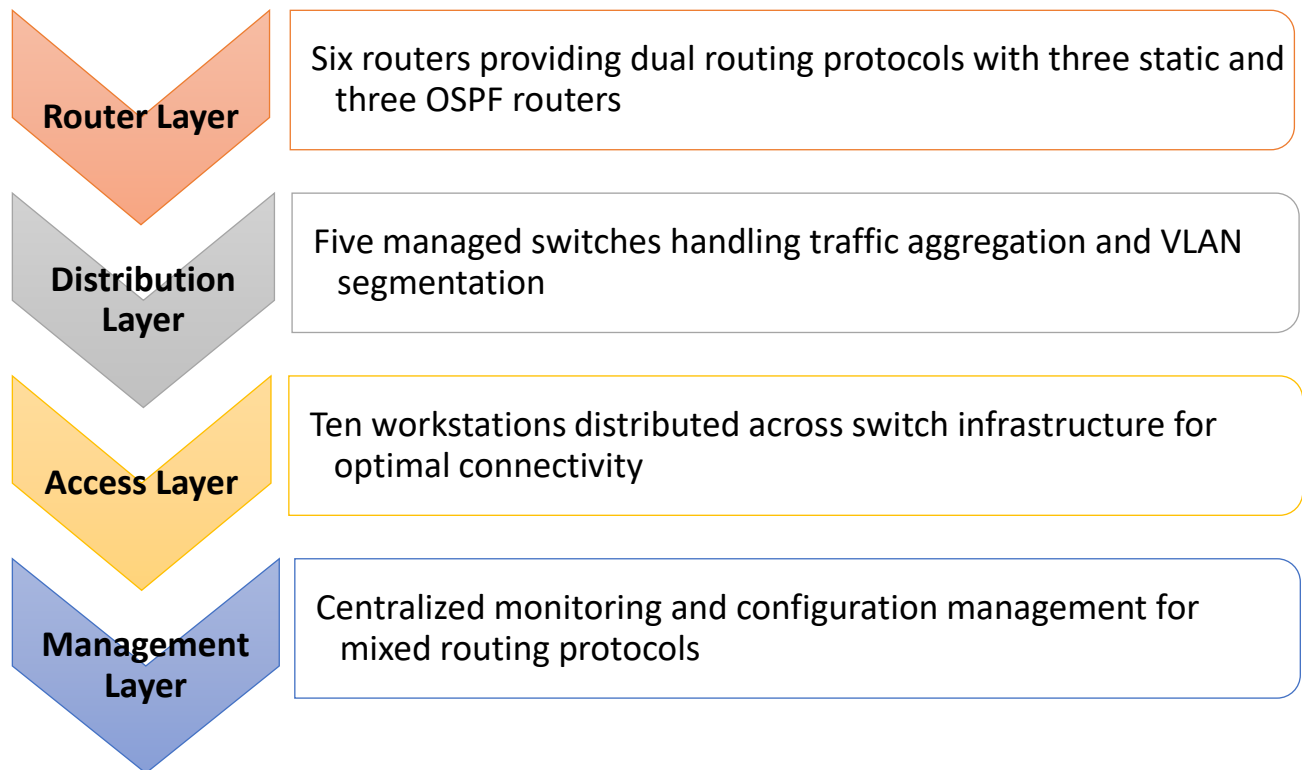
- The network is divided into five VLANs (10, 20, 30, 40, 50).
- There are 6 routers (R1–R6) connected in a hierarchical manner to provide inter-network connectivity.
- 5 switches are used for VLAN segmentation and device access.
- FTP and Web servers are placed in VLAN 50, while users are distributed across VLANs 10–40
- There are 2 access controller and 2 access points providing a wireless communication connection for 2 cell phones.

Device Naming Convention

Device Type	Name in Diagram	Count
Routers (R)	R1, R2, R3, R4, R5, R6	6
Switches (SW)	SW1, SW2, SW3, SW4, SW5	5
Access Controllers (AC)	AC3, AC4	2

Access Points (AP)	AP1, AP2	2
Servers	FTP Server, Web Server	2
End Devices	PC1, PC2, PC4, PC5, PC6, PC7, PC8, PC9, Client1, Client2, Client3, Cellphone1, Cellphone2	13

Network Hierarchy Architecture



VLAN and IP Addressing Scheme

The network is segmented into five VLANs, each with a dedicated subnet and a VRRP-configured gateway.

VLAN ID	VLAN Name	Network Address	Gateway (VRRP VIP)	Device Assignments
VLAN 10	PC/Client Access	192.168.3.0/24	192.168.3.1	PC1, PC7, Client1

VLAN 20	PC/Client Access	192.168.4.0/24	192.168.4.1	PC2, PC6, Client2, Cellphone 2
VLAN 30	PC Access	10.200.1.0/24	10.200.1.1	PC5, PC8, Cellphone1
VLAN 40	PC Access	10.150.1.0/24	10.150.1.1	PC4, PC9, Client3
VLAN 50	Server Farm	192.168.50.0/24	192.168.50.1	FTP Server, Web Server

Detailed IP Address Assignments

Device	VLAN	IP Address	Gateway
PC 1	10	192.168.3.100	192.168.3.1
PC 7	10	192.168.3.101	192.168.3.1
Client 1	10	192.168.3.102	192.168.3.1
PC 2	20	192.168.4.100	192.168.4.1
PC 6	20	192.168.4.101	192.168.4.1
Client 2	20	192.168.4.102	192.168.4.1
PC 5	30	10.200.1.100	10.200.1.1
PC 8	30	10.200.1.101	10.200.1.1
Client 3	40	10.150.1.102	10.150.1.1
PC 4	40	10.150.1.100	10.150.1.1
PC 9	40	10.150.1.101	10.150.1.1
FTP Server	50	192.168.50.100	192.168.50.1
Web Server	50	192.168.50.101	192.168.50.1

Routing Design

The network employs a Mixed Routing Design to leverage the strengths of both static and dynamic routing protocols.

North Routers (R1, R2, R3) - Static Routing Domain

These routers form the core of the network and use Static Routing for predictable, deterministic traffic paths. This is ideal for the backbone where route changes are infrequent and control over traffic flow is paramount.

Router	IP Address	Connected Routers
R1	10.100.2.4	R3

	192.168.2.3	R2
R2	192.168.2.2	R1
	10.100.3.3	R3
R3	10.100.3.2	R2
	10.100.2.3	R1

South Routers (AR4, AR5, AR6) - OSPF Routing Domain

These routers connect the Distribution Layer switches and use OSPF (Open Shortest Path First) for dynamic, adaptive routing. OSPF provides automatic topology discovery, fast convergence, and load balancing, which is crucial for the access and distribution layers where network changes and failures are more likely.

Router	IP Address	Connected Routers
R4	192.168.30.3	R6
	192.168.40.3	R5
R5	192.168.40.2	R4
	192.168.60.2	R6
R6	192.168.60.3	R5
	192.168.30.2	R4

Routing Integration

The two routing domains are connected via the Distribution Layer switches and their respective uplink interfaces on the routers. This strategic combination ensures optimal traffic distribution while maintaining network stability through redundant pathways and protocol-specific advantages.

First Hop Redundancy (VRRP) Configuration

VRRP is configured on the VLAN Interfaces (Vlanif) of the Distribution Layer switches to provide a highly available default gateway for all end devices. The VRRP Virtual IP (VIP) is the default gateway for the corresponding VLAN.

VRRP Master/Backup Assignment

The VRRP configuration from the draft document suggests a primary/secondary setup, likely across SW1/SW2 and SW4/SW5, with one

switch acting as the master (higher priority) and the other as the backup for each VLAN.

VLAN ID	VRRP VIP	Master Switch (Example)	Backup Switch (Example)
VLAN 10	192.168.3.1	SW1 (Priority 120)	SW4 (Priority 100)
VLAN 20	192.168.4.1	SW2 (Priority 120)	SW5 (Priority 100)
VLAN 30	10.200.1.1	SW4 (Priority 120)	SW1 (Priority 100)
VLAN 40	10.150.1.1	SW5 (Priority 120)	SW2 (Priority 100)
VLAN 50	192.168.50.1	SW3 (Single Point)	N/A

Security Framework Implementation

1. Access Control Lists (ACLs)

ACLs are implemented to control traffic flow and enforce security policies between different VLANs and network segments.

- **Inter-VLAN Security:** ACLs are applied to the VLAN interfaces on the Distribution Layer switches to filter traffic between user VLANs (10, 20, 30, 40) and the Server Farm (VLAN 50).
- **Protocol Security:** ACLs are used to restrict access to management protocols (e.g., Telnet, SSH) and to filter unauthorized routing updates.

2. Wireless Security

The network includes two Access Controllers (AC3, AC4) and two Access Points (AP1, AP2) to provide secure wireless connectivity for mobile devices (Cellphone1, Cellphone2).

- **802.1X Authentication:** Port-based 802.1X authentication is implemented on the access switches to ensure only authorized devices can connect to the network.
- **VLAN Assignment:** Wireless clients are dynamically assigned to their respective VLANs (VLAN 20 for Cellphone2, VLAN 30 for Cellphone1) upon successful authentication.

Implementation Roadmap

